

FILE COMPRESSOR REPORT

Project Title: File Compression Tool using Huffman Coding

Objective:

To develop a universal file compression and decompression system using Huffman Coding in C++, capable of handling any file type including text, images, PDFs, etc., to reduce storage space and ensure data integrity.

Technologies Used:

- C++
- STL (Standard Template Library)
- File Handling (binary I/O) - Huffman Coding Algorithm

Project Description:

This project implements Huffman Encoding to compress any type of file by calculating frequency of bytes and assigning optimal binary codes. The compression process involves:

- Reading input file as binary
- Constructing frequency map and Huffman tree
- Writing encoded data with a custom header that includes dictionary and padding info
- Bit-level compression of file content

Decompression involves reading the header dictionary, decoding bits back to characters, and restoring the file.

Features:

- Universal file type support (text, image, PDF, binary, etc.)
- Efficient memory usage with STL data structures
- Bit-level manipulation using bitsets for actual compression
- Dictionary stored in output file for decoding
- Works via command line interface with user input for file paths

Limitations:

- Compression may not always be effective on already compressed files (e.g., .zip, .jpg, .mp4) - Overhead due to dictionary storage for small files . Sometimes some files might not be compressed due to Huffman dictionary overhead, bit-padding to byte alignment, lack of compressible patterns(mainly occurs in binary files like PDFs, images, videos etc. have **high entropy (nearly uniform frequency)**).

Conclusion:

The Huffman-based universal compression tool offers a strong understanding of data encoding, binary operations, and system-level file handling, making it an ideal systems/DSP/DSA-focused resume project.

Future Scope:

- Add GUI interface
- Add support for batch compression
- Integrate other compression algorithms like LZW or Run-Length Encoding